PureLink

For UN SDGs Goal 6: Clean Water and Sanitation

Project
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Individual Project | Product Design | Service Design

Brief

In response to health and public perception concerns arising from domestic water pollution in Beijing, I conducted cultural probe and co-creation group research. This culminated in the development of **a service process system incorporating a simple water purifier and accompanying app** achieving the design objective of enabling residents to uniformly access purified water.

Duration

2025/7/27-2025/10/8



Desktop Research

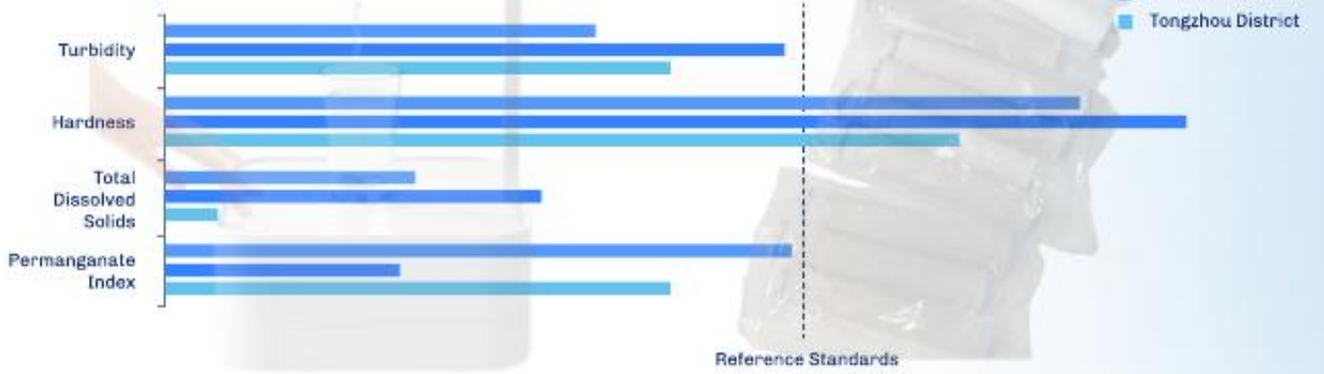
Why does domestic water pollution occur?



Finding Domestic water pollution is commonplace in older residential estates in Beijing. Residents are currently taking spontaneous measures to address water pollution.

What are the pollution indices?

Water Quality Indicator Test Results in Beijing in the Second Quarter of 2025 (Partial)



Current Solutions Are Poor:

Residents are taking independent measures to address domestic water pollution.



Install water purifiers/ Purchase bottled water

Cons
- Increased economic costs
- Tackling drinking water contamination alone is costly and difficult for daily use

Make and purchase items for testing and purifying water quality

Cons
- Inequality in access to water quality information
- Lack of capacity for accurate water quality testing
- Inappropriate purification methods may adversely affect physical health

Stakeholder Map



Problem

Insoluble issues ✕
- Pipework issues in older housing estates **cannot be resolved immediately**.
- Official investigations into incidents are protracted and proceed at a sluggish pace.

Solvable issues ✓
- The property management and waterworks **engage in mutual buck-passing.**
- Rumours have caused **psychological panic** among residents.

User Research

Culture probe



- a notebook for recording experiences
- a pen
- some sticky notes
- several tasks *(with notes on moments when I wanted them to record their water usage)*
- a water quality testing pen
- a mobile phone for taking photos

I distributed this toolkit to two community property managers and three neighbours.



We can't solve residents' issues directly — we can only tell them to wait for updates at home.

Property management staff answered the residents' complaint telephone line



A water tester is too expensive for me, and it can't purify the water, so I find it troublesome.

After receiving the toolkit, neighbour Xiaoqin began recording and monitoring her daily water usage.

Interview

In cases of domestic water pollution, what particularly challenging difficulties have you encountered?

Much of the water pollution stems from ageing pipe networks, which ***fall outside our remit***. Yet residents persistently blame us.

Property Management

Following contamination incidents, the council issues notices instructing us to seek resolution through the property management. The management then ***deflects responsibility onto the waterworks.*** Ultimately, ***no one steps in to address the issue.***

Resident

Persona



Ruoxi

Age: 26
Area: Haidian District, Beijing

Situations

- Residing in an older housing estate **(built over 20 years ago)**
- Water coming out of various taps in the home looks **yellowish and gives off an unpleasant smell** (especially during the rainy season)

Needs

- Report contamination to property management but **get no or slow replies**
- Spread rumours online, fueling **panic and distrust**
- Notice pollution **only when water smells, discolours, or causes illness**
- Install filters or buy bottled water, **increasing costs**
- Rely on bottled water but **still face exposure when washing or bathing**
- Seek independent testing **but struggle to access credible reports**

Summary

Residents

Lack of clear information fuels rumours and distrust
Contamination is only noticed when severe
Filters and bottled water add costs but can't replace daily use
Reliable third-party testing is rare and inaccessible

Property Management

Estate pipe issues cause contamination and are hard to fix without support
Repairs need funding, slowing progress
Poor government communication delays updates
Residents are unclear who should handle testing or backup supply
Resident complaints are frequent

Government

Plant water meets standards, yet resident complaints persist, eroding trust

Pain Point Classification

Physical health

Awareness only when pollution becomes severe and evident
Psychological burden from prolonged suspicion and unease

Financial resources

Residents using filters or bottled water face higher costs.
Property management can't afford pipe repairs or cleaning costs

Trust

When pollution occurs, social media rumours erode government trust
The property management and waterworks blame each other instead of fixing the problem, reducing residents' trust
Residents doubt government water quality data, even though it meets standards

Communication channels

Residents want trusted third-party testing, but credible, public results are rare and hard to access
During pollution events, official notices are slower than social media, and technical terms are hard to understand

Accountability

Residents received no help during pollution, as property management and waterworks blamed each other
Residents don't know who to hold accountable, since the pollution source—factory, main sewer, or residential pipes—is unclear

Co-Design







Government Staff
Property Management
Residents

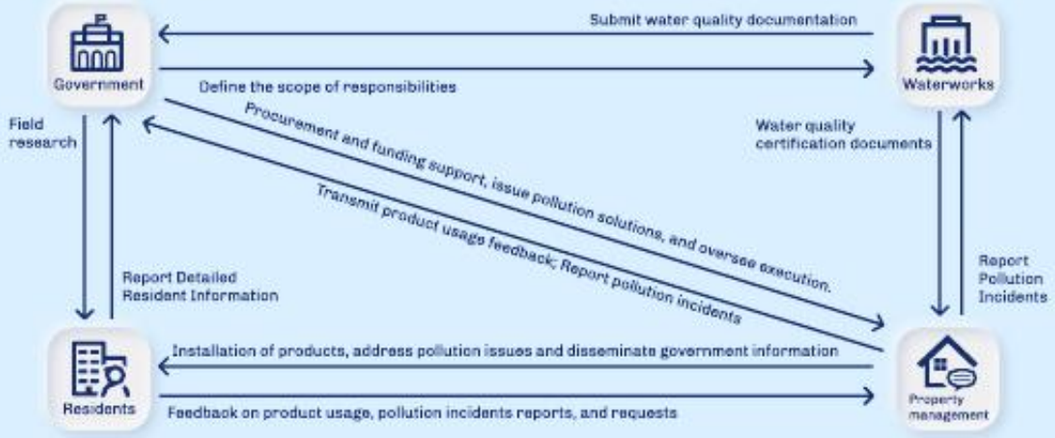
Outcomes

- Use products that **show water quality clearly and purify effectively.**
- Property management offers **low-cost systems to ease expenses.**
- **Publicly share results** from third-party testing.
- Government and property give **timely, clear updates through a shared platform.**
- **Define responsibilities** for all parties during pollution response.

Concept

How Might We	addressing water safety and information credibility communication issues
For	residents
So That	cross-sector collaboration can **ensure equitable, resilient, and trustworthy access to safe water** for all residents

System Map



Design value

Multi-stakeholder collaboration platform:
A unified system connecting key parties to **improve communication and prevent blame-shifting.**

Participatory Governance:
Enabling residents to voice opinions and **enhance their influence in public services.**

Lowering Access Barriers:
Property management provides shared equipment, allowing wider service access **at low cost.**

Technology Meets Design:
Combining purifiers and an app to make water data easy to understand.



Interfaces

Resident Interfaces

Home





Adjust the filtration mode / Real-time Details / Real-time status

Notices



Query Documents / Contact Support

Online Mall



Purchase consumables / Request Assistance

My



View Product Manual / Submit Information

Property Management Interfaces

Online Mall



Centralized procurement / Request Assistance

Notices



Submit Inndents / Contact Support

My



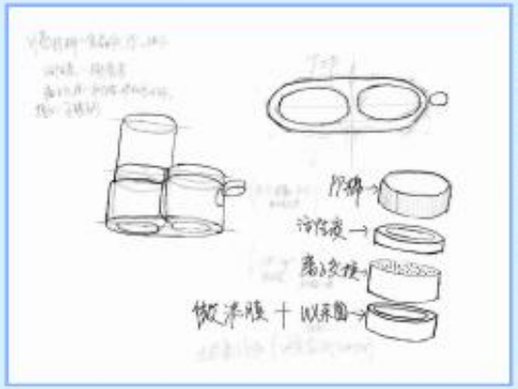
View Product Manual / Submit Information

User Tests



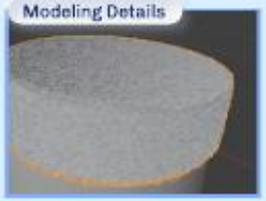
Product Design

Sketches



Purelink's product is an **affordable portable water purifier** for domestic use. It can be installed on various taps to provide purification, monitor water quality, and transmit data to a mobile application.

Modeling Details



Materials Testing

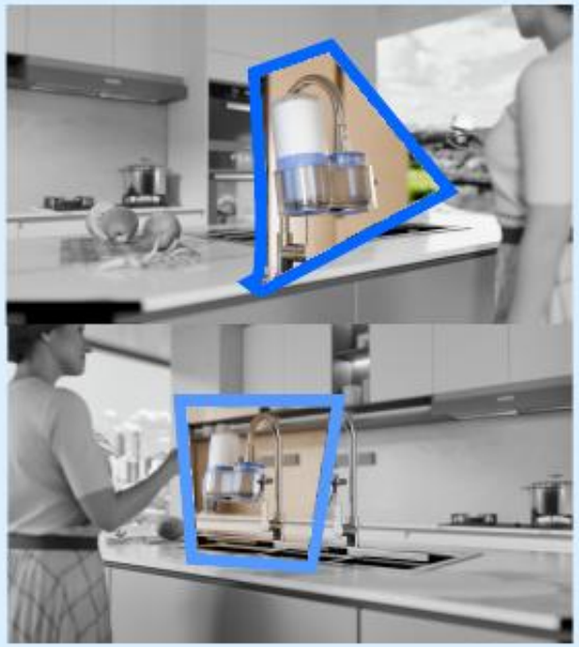


Final Work



User Scenario

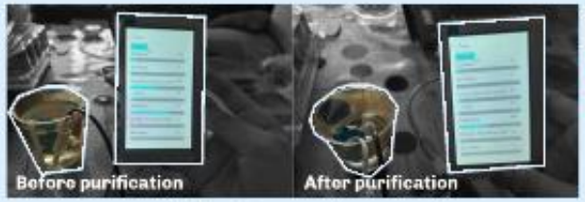
In this scenario, residents install the water purifier on the kitchen tap outlet and **set it to the direct drinking water mode**, enabling them to **obtain water suitable for immediate consumption.**



Procedure



Results



*a **significant decrease** in parameters*

Coding

01 Ultraviolet lamp

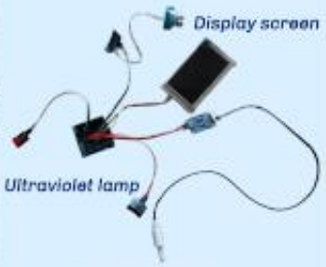


*Activates the ultraviolet lamp **when water flow is detected***



*Burn the program onto the **Arduino UNO board***

02 Water Quality Testing Sensors





Service Blueprint



User Journey	Noticing the water is cloudy Call the property management	Download the app as arranged by the property management	Submit resident details	Arrange for property management to install the product on-site	Learn how to use the product and app	Browsing and posting on social media, identifying rumours View real-time water quality data and the water quality factory documentation	Install the product at different outlets	Submit feedback and suggestions	Report any issues encountered during use to the product customer service team	Accept government field research	Purchase and replace consumables and accessories independently	Obtain clean domestic water
Front Stage	Awaiting the outcome of the incident investigation Property switchboard operator	Property Management staff Mobile devices App	App interfaces	Installation personnel Product	Property management staff Product App	App interfaces Social media interfaces Information release platform interfaces	Product	App interfaces	App interfaces	Government officials	App interfaces	
Back Stage	Water Supply Maintenance Technician	App Development Specialist App code	Property Information Collector Information System Code Information Data	Product Manufacturer Product Development Specialist Water Purification Technology	After-sales Service Training Content code	Water quality sensor News Producer Incident Investigator Media user Data	Consumables Product Development Personnel	Information Receiving Personnel App communication code	Product Customer Service Representative App communication code	Planner Head of Investigation	Online shopping system code Product supplier	
Support	Telephone System			Government policy support Government funding support		Government Public Management	Water supply system	Government oversight and administration			Logistics system	

Line of Interaction

Line of Visibility